CADILLAC FORMULA 1

Graduate Aero Mechanical Design Engineer

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16-hour action plan • 14 focused modules • practice questions and answer checkpoints

Prepared 18 October 2026. Use current official team and FIA information immediately before your interview.

What this pack can and cannot do

Understanding and practising this material should put you in a much better position for both technical and general questions. It cannot guarantee progression: the panel may introduce a CAD exercise, drawing test, unexpected problem or different emphasis. Prioritise explaining decisions and asking good questions over reciting answers.

The most important adjustment

This is an aero-mechanical design opportunity supporting a wind-tunnel model and testing capability, not a pure CFD vacancy. Your CFD expertise is a strength, but mechanical reasoning, CATIA, drawings, manufacture, interfaces, measurement, teamwork and genuine interest in F1 deserve at least as much preparation.

Exactly 16 hours

The sections total 15 hours of active work. Add 60 minutes of breaks: six 10-minute breaks spread across the programme. Suggested two-day split: sections 1–7 total 8 hours 15 minutes active; sections 8–14 total 6 hours 45 minutes active. Add three breaks each day. Split into more days if preferable; sleep and recall are more useful than an exhausted cram.

How to use each section

Read the notes once, complete the actions, then answer the practice questions aloud with notes closed. Compare against the checkpoints. Mark red if you cannot explain it; amber if prompted; green if you can sketch, explain and defend it. Redo red items before adding optional reading.

Personal evidence and accuracy warning

Several earlier draft answers were too broad. Do not claim personal execution of all group FEA, a demonstrated CFD pressure-map transfer, formally completed test campaigns or proven causes of discrepancies unless your records support them. Do not assume model hardware examples are confirmed Cadillac assignments. Do not claim a new interest is lifelong fandom.

No fabricated experience

Do not learn scripts that imply you have done composite lay-up, CNC machining, PLM release, advanced GD&T or specific model-system work if you have not. Explain transferable experience and the supervised learning you need. Confidentiality applies even to a thesis you have uploaded for assistance.

Source use

This is a preparation guide, not proprietary Cadillac information. Public role/source references are labelled S1–S9; project evidence is P1–P3. Engineering notes are foundational revision and example reasoning, not released design instructions. Verify actual design loads, standards, materials and safety requirements with the employer.

Your action plan

1. Understand the job: model hardware, not a pure CFD post — 45 minutes
2. F1 and Cadillac: genuine interest in the sport — 75 minutes
3. Aero fundamentals: wings, floor and diffuser — 75 minutes
4. Wind-tunnel testing, scaling and correlation — 75 minutes
5. Mechanical design workflow and CATIA — 90 minutes
6. Mechanics, stiffness and hand calculations — 75 minutes
7. FEA and CFD-informed structural assessment — 60 minutes
8. Materials and manufacturing selection — 90 minutes
9. GD&T, fits, inspection and tolerance stacks — 60 minutes
10. Mechanisms, fasteners, actuation and instrumentation — 60 minutes
11. CFD: defend your thesis without turning this into a CFD interview — 60 minutes
12. Your project evidence and ownership — 45 minutes
13. General interview and motivation answers — 45 minutes
14. Integrated mock interview and final readiness — 45 minutes

Finished-work checklist

[ ] One 90-second introduction and one honest why-Cadillac answer.

[ ] One current F1 sporting observation you can discuss naturally.

[ ] Four project evidence cards with exact personal ownership.

[ ] One labelled design sketch, one tolerance stack and three hand estimates.

[ ] One manufacturing-selection matrix and one FEA verification checklist.

[ ] One short thesis explanation with method distinctions and limitations.

[ ] One recorded mock interview, reviewed against the readiness score.

[ ] Three questions for the panel and confirmed interview logistics.

Optional extra time - only after the core plan

If you can spend more than 16 hours, first do a second 45-minute mock, then 45 minutes of CATIA/drawing practice and 30 minutes reviewing current F1 coverage. Do not prioritise obscure technical trivia, detailed regulations or extra CFD literature over weak mechanical-design answers.

1. Understand the job: model hardware, not a pure CFD post

45 minutes | Outcome: Explain what the role does, why it matters and why you want it.

Actions - produce something, do not just read

[ ] Re-read the vacancy and highlight model systems, electromechanical design, CATIA, FEA, manufacturing and instrumentation.

[ ] Draw the development loop: aero concept / CFD -> model design -> manufacture / inspection -> tunnel test -> correlation -> design decision.

[ ] Write two sentences distinguishing aerodynamic shape design from aero-mechanical design.

Review notes and reasoning

Confirmed versus inferred

The public vacancy describes developing the 60% wind-tunnel model and testing facility, including mechanical and electromechanical systems. It does not establish your exact allocation, proprietary tools, hands-on time or CFD workload. Brackets, mechanisms, fixtures and instrumentation integration are plausible examples, not confirmed assignments. [S1]

What a day might look like

A senior engineer gives you a defined hardware problem. You clarify loads, interfaces and packaging, sketch alternatives, model a part/assembly, check stiffness and clearances, review with colleagues, produce a drawing and support manufacture or a build issue. Some days may be almost entirely CAD and checking. Do not assume daily CFD or tunnel operation.

Translate your motivation

Your CFD experience helps you understand why geometry, stiffness and test repeatability matter. The job still requires enthusiasm for mechanical design, drawings, manufacturing and collaboration. Decide honestly whether you want these activities; do not promise a design career only to gain access to CFD.

Example problem

A pressure-sensor mount must fit inside the model without obstructing tubing. You clarify measurement position, stiffness, space, attachment, access, cable bend radius, mass and replacement needs. Then compare concepts, check loads, define tolerances, arrange review and verify the installed position.

Practice questions - close your notes first

1. What would you actually do in this job?

Answer checkpoints: Explain model hardware and test capability, CAD-to-manufacture delivery and review; distinguish this from designing the tunnel building or being the lead aerodynamicist.

2. Why not a pure CFD job?

Answer checkpoints: Give your genuine interest in hardware/test work, backed by one specific rig example. Acknowledge that this is design-led.

3. What would you ask the hiring manager?

Answer checkpoints: Ask about model versus facility work, typical graduate projects, design release and mentoring.

Exit check: Can I explain this simply, sketch it if needed, and answer one follow-up without a script? RED / AMBER / GREEN

2. F1 and Cadillac: genuine interest in the sport

75 minutes | Outcome: Address Judy's feedback with real engagement, not memorised fandom.

Actions - produce something, do not just read

[ ] Read the official Cadillac team profile and three recent team news items. Note date, source and one observation each.

[ ] Watch one recent race highlights package and one qualifying highlights package from official F1 coverage.

[ ] Explain the current race weekend and why qualifying, tyres, pit timing and reliability affect results.

[ ] Write one sporting storyline and one team-development question you genuinely want to follow. Refresh facts the day before interview.

Review notes and reasoning

What to know

Know the drivers, current leadership, debut-season context, recent performance and next event from official sources. Cadillac's announced debut drivers were Valtteri Bottas and Sergio Perez. Recheck present details rather than relying on an old interview script. [S2, S3]

Sporting foundations

Practise explaining qualifying versus race pace; the basic purpose of tyre compounds and pit stops; drivers' and constructors' championships; safety-car effects; and the role of reliability. Check current weekend and sporting rules, particularly sprint formats, against official F1/FIA explainers rather than assuming every weekend is identical.

Authentic motivation

A truthful line is: My interest began through the engineering, and I have started following the sport more closely. What interests me is seeing engineering choices turn into performance over a race weekend. Add an actual observation from something you watched, not a claim of lifelong support.

What counts as evidence of interest

Explain what happened in the coverage you watched and what it made you curious about. You do not need to admire every driver or learn decades of statistics. Do not frame the sport as incidental to obtaining an engineering job.

Practice questions - close your notes first

1. Have you always followed F1?

Answer checkpoints: Be honest about when you started. Describe what you have actually watched/read and why it caught your attention.

2. Why Cadillac rather than any engineering employer?

Answer checkpoints: Connect the team's development context, role-specific work and a current verified observation. Avoid generic superlatives.

3. What interested you in the last race you watched?

Answer checkpoints: Describe one real event, its performance/strategy implication and a thoughtful question. Do not invent a race anecdote.

Exit check: Can I explain this simply, sketch it if needed, and answer one follow-up without a script? RED / AMBER / GREEN

3. Aero fundamentals: wings, floor and diffuser

75 minutes | Outcome: Explain the physics simply and identify implications for model hardware.

Actions - produce something, do not just read

[ ] Sketch a wing section with pressure forces, a floor side view and an expanding diffuser.

[ ] Give a 60-second explanation of downforce, drag and separation without equations.

[ ] Explain why ride height, yaw and deformation could change a measurement.

[ ] Read an official current-rule aero explainer: do not memorise previous-generation layouts as current Cadillac architecture.

Review notes and reasoning

Pressure, downforce and drag

Aerodynamic force is the integrated effect of pressure and wall shear over the surface. Downforce is the downward component; drag opposes motion. Downforce can improve tyre loading but is not free of drag or other trade-offs. Lift/downforce magnitude scales approximately with dynamic pressure times reference area times coefficient.

Floor / ground effect

The floor is the underside aerodynamic surface. Its interaction with the nearby ground changes pressure and flow under the car. A low-pressure underfloor can generate downforce. Do not say any smaller gap always gives more downforce: excessive restriction, separation and configuration sensitivity can reverse the trend.

Diffuser

An expanding passage towards the rear helps slow underbody flow and recover pressure. Geometry and downstream conditions influence upstream mass flow and low-pressure performance. Too aggressive a pressure recovery can cause separation; a diffuser does not create unlimited suction.

Boundary layers and separation

Viscosity produces a near-wall velocity gradient. An adverse pressure gradient asks the flow to move towards higher pressure; a low-momentum boundary layer can separate. This changes force and wake behaviour. Vortices can redistribute momentum and influence leakage or separation, but can also create losses.

Model relevance

A small deflection or angle error can change the tested geometry. Floor clearance, flap position and component alignment must be measured and repeated. Wheel wakes and cooling flows are relevant aero topics, but not proof that this graduate personally designs every such system.

Practice questions - close your notes first

1. Explain ground effect to a non-engineer.

Answer checkpoints: Low pressure from floor-ground flow interaction pushes the car down relative to pressure above. Include clearance sensitivity.

2. Why might a diffuser stall?

Answer checkpoints: Adverse pressure gradient and insufficient near-wall momentum; relate to geometry and operating conditions.

3. Does a stiffer wing always make the fastest car?

Answer checkpoints: Separate model fidelity, legal compliance, structural safety and aerodynamic optimisation; requirements determine the answer.

4. What happens to aerodynamic force if speed doubles?

Answer checkpoints: With unchanged density, area and coefficient, dynamic pressure and force increase fourfold. State assumptions.

Exit check: Can I explain this simply, sketch it if needed, and answer one follow-up without a script? RED / AMBER / GREEN

4. Wind-tunnel testing, scaling and correlation

75 minutes | Outcome: Understand what the model measures and why precision matters.

Actions - produce something, do not just read

[ ] Draw the model, moving road, support, force balance, pressure tubing and position sensors as a conceptual diagram.

[ ] Explain Reynolds-number mismatch and why a 60% model is not automatically dynamically identical to a full car.

[ ] Write a test-readiness checklist and a CFD-versus-test discrepancy checklist.

Review notes and reasoning

Scale and restrictions

A 60% model means linear dimensions are 0.6 times full size, not 60% of volume or mass. The cited FIA operational regulations cap restricted testing model scale at 60% and air speed at 50 m/s. Regulations and allocations must be rechecked for the interview date; the caps do not mean the tunnel always runs at that speed. [S4]

Scaling

Reynolds number Re = rho U L / mu compares inertial and viscous effects. At the same fluid properties and speed, reducing length to 0.6 reduces Re to 0.6. Matching force coefficients does not remove every scale, roughness, deformation or tyre issue. Geometrically similar volume scales as length cubed; actual model mass need not follow that law.

Moving ground

A moving belt better represents the road's motion relative to the car in a car-fixed frame. Wheels may rotate. Match relative wall velocities carefully in CFD; a stationary floor can give a different boundary layer and underbody flow.

Measurements

Force/moment balances measure loads; pressure taps sample local pressure. Position sensors verify configuration. Calibration, zeroing, drift, mechanical interference, tube leaks and temperature effects can influence measurements. Repeatability means repeated results agree under controlled conditions; it is not identical to accuracy or validation. NPL emphasises reference-based measurement and uncertainty. [S5]

Correlation

If CFD and tunnel disagree, first check geometry/version, reference areas, coordinate/sign conventions, speed, attitude, wheel/ground treatment and data processing. Then inspect supports, blockage, turbulence, transition, mesh/time resolution, deformation and instrumentation. Never tune a model until it matches a number without physical justification.

Practice questions - close your notes first

1. Why not test a full-size car?

Answer checkpoints: Discuss practical resources and regulatory scale limits without claiming scale testing has no disadvantages.

2. Why can model and full-size results differ?

Answer checkpoints: Re, surface/detail scaling, tyre shape, support interference and deformation; distinguish these mechanisms.

3. A new floor gives a small improvement: would you trust it?

Answer checkpoints: Compare against measurement uncertainty, repeatability, baseline drift, configuration checks and repeat runs.

4. How would you troubleshoot unexpected balance forces?

Answer checkpoints: Check zeros, calibration, load paths, interference, coordinate system, operating conditions and safe rerun plan.

Exit check: Can I explain this simply, sketch it if needed, and answer one follow-up without a script? RED / AMBER / GREEN

5. Mechanical design workflow and CATIA

90 minutes | Outcome: Walk through a real component from requirement to released drawing.

Actions - produce something, do not just read

[ ] Choose a non-confidential fixture and sketch its interfaces, load path and assembly sequence.

[ ] Rebuild a simple parametric bracket/assembly in CATIA if available; otherwise sketch the feature tree and constraints.

[ ] Prepare a two-minute CAD example including a design change and why it was needed.

[ ] Make a one-page design review sheet: requirements, concepts, checks, manufacture, inspection and residual risks.

Review notes and reasoning

Interfaces in plain English

An interface is where your part meets something else: a shaft bore, mating face, bolt pattern, sensor connector or cable route. Define fit, location, loads, allowable movement, access and ownership at each connection.

Workflow

Clarify function and measurable requirements -> define loads and interfaces -> sketch concepts -> compare -> CAD -> hand calculations/FEA -> tolerancing and process review -> peer check -> release -> manufacture/inspection -> assembly/test -> controlled revision. Ask for missing inputs; do not guess safety-critical requirements.

Parametric CAD

Use design intent: sensible datum planes, named features, dimensions driving function and stable references. Avoid fragile chains of references where practical. Check clearances, tool access, assembly order and whether changes propagate correctly. Explain actual CATIA workbenches used; V5 experience is not proof of 3DEXPERIENCE/PLM competence.

Drawing and release

Communicate material, finish, key dimensions, tolerances, datums, threads and applicable notes. Include revision control and a bill of materials as needed. Confirm approval and change-control process; do not alter a released design informally.

Design trade-off example

A monolithic housing may be stiff but obstruct access or be heavy. A modular housing may improve assembly and replacement but add interfaces, tolerance stack-up and fastener risks. Use the functional requirement rather than assuming fewer or more parts is always better.

Practice questions - close your notes first

1. Design a sensor bracket for a crowded model.

Answer checkpoints: Ask about loads, location accuracy, stiffness, space, wiring, environment, assembly and inspection; then sketch concepts.

2. How do you ensure your CAD is easy to change?

Answer checkpoints: Design intent, parameters, reference strategy, clean feature tree, checks after changes and documented interfaces.

3. What goes on a manufacturing drawing?

Answer checkpoints: Function-led dimensions/tolerances, material, finish, datums, threads, revision and checking information.

4. What is your CATIA limitation?

Answer checkpoints: State a genuine gap and the concrete way you would close it; do not bluff software experience.

Exit check: Can I explain this simply, sketch it if needed, and answer one follow-up without a script? RED / AMBER / GREEN

6. Mechanics, stiffness and hand calculations

75 minutes | Outcome: Use first principles before a simulation and explain assumptions.

Actions - produce something, do not just read

[ ] Sketch a cantilever with end force, a bolted bracket and a shaft under torque. Draw reactions.

[ ] Work the three practice estimates below on paper, with units.

[ ] Explain strength, stiffness, fatigue and buckling as different checks.

Review notes and reasoning

Core relationships

Stress = F/A for simple uniform axial loading; axial extension = FL/(EA). For a slender linear-elastic cantilever with a small transverse end load: maximum bending moment FL and tip deflection FL^3/(3EI). Bending normal stress = My/I. These are idealised beam relations, not universal formulas for every bracket.

Geometry matters

Young's modulus E characterises elastic material stiffness; second moment of area I captures cross-section geometry in bending. For a rectangular section bending about its centroidal width axis, I = b h^3/12. Increasing depth in the correct direction is often much more effective than adding width.

Different failure modes

Yield is permanent deformation for a ductile material. Buckling is an instability influenced by geometry, restraints and imperfections. Fatigue is damage under cycling. A part can be below yield yet too flexible, vibration-sensitive or fatigue-limited. Composite failure requires an appropriate directional/laminate assessment rather than only an isotropic yield comparison.

Load path

Trace force from component to bracket to fasteners to supporting structure. Check eccentricity, bending, bearing, local contact and restraint assumptions. A safety factor does not repair a missing load case or incorrect boundary condition.

Dynamics

Natural frequency depends on stiffness and mass; a simple oscillator has f = (1/2pi) sqrt(k/m). Excitation near a natural frequency may amplify vibration depending on damping. Increasing stiffness can raise natural frequency; adding mass can lower it. Assess the actual system, not just the isolated part.

Practice questions - close your notes first

1. Estimate deflection: F=10 N, L=0.10 m, E=70 GPa, I=1e-9 m^4.

Answer checkpoints: FL^3/(3EI) = 4.76e-5 m, about 0.048 mm. State beam/support assumptions and whether the requirement permits it.

2. If cantilever length doubles, what happens?

Answer checkpoints: Under the same end load and section, deflection rises eightfold and maximum bending moment doubles.

3. If rectangular-section depth doubles?

Answer checkpoints: At unchanged width, I increases eightfold; elastic bending deflection reduces to one eighth under the same idealised load.

4. A part passes stress but fails clearance in the tunnel. Why?

Answer checkpoints: Deflection, thermal growth, assembly tolerance, wrong load or restraint assumptions. Strength alone is insufficient.

Exit check: Can I explain this simply, sketch it if needed, and answer one follow-up without a script? RED / AMBER / GREEN

7. FEA and CFD-informed structural assessment

60 minutes | Outcome: Explain trustworthy structural analysis and your exact contribution.

Actions - produce something, do not just read

[ ] Revisit your structural report and identify what you personally modelled, solved and interpreted.

[ ] Write actual load source, supports, material assumptions, mesh study and limitations on one sheet.

[ ] Explain one-way coupling without calling every pressure-loaded FEA a fully coupled CFD-FSI analysis.

Review notes and reasoning

Model ingredients

FEA represents geometry, materials, mesh, loads, contacts and constraints. Mesh quality and refinement affect accuracy and convergence, but a refined mesh cannot fix unphysical restraints. [S6]

Trust checks

Start with free-body diagrams and load estimates; check units, reactions/equilibrium, deformation shape, contact and load transfer, mesh sensitivity and output relevance. Compare to simple theory or test. Assess whether plasticity, large deformation, fatigue or dynamics matters.

Stress singularities

Ideal point loads, sharp corners and perfectly fixed edges can produce mesh-dependent peaks. Identify whether a peak is numerical or a real stress concentration. Improve physical modelling and evaluate appropriate quantities; do not simply delete inconvenient stress values.

One-way coupling accuracy

CFD-derived loads applied to structure with no deformation fed back to flow are one-way coupled. Two-way coupling updates both directions. If the report used an assumed pressure rather than an actual mapped CFD field, say pressure-loaded FEA informed by the fluid analysis. Check your files before choosing the stronger description.

Ownership

A group report is not proof you personally performed all FEA. Use contributed to, supported or collaborated on where accurate. Discuss your own decisions, distinguish team output, and do not claim the static material comparison proved the failed shaft safe.

Practice questions - close your notes first

1. How would you verify a bracket FEA?

Answer checkpoints: Loads/reactions, analytical comparison, support sensitivity, mesh study, realistic contacts and deflection as well as stress.

2. Why might stress increase as the mesh is refined?

Answer checkpoints: Resolved gradients or singularity; distinguish convergence of useful quantities from a non-convergent idealised peak.

3. When is one-way fluid-structure treatment appropriate?

Answer checkpoints: When deformation feedback is small for relevant flow outputs; justify, do not assume.

4. What did you personally do in the group FEA?

Answer checkpoints: Give exact ownership, load derivation and limitations; do not recite a team result as your individual work.

Exit check: Can I explain this simply, sketch it if needed, and answer one follow-up without a script? RED / AMBER / GREEN

8. Materials and manufacturing selection

90 minutes | Outcome: Compare processes and materials against requirements, not prestige.

Actions - produce something, do not just read

[ ] Build a selection matrix for a precision bracket, aero surface and temporary instrument housing.

[ ] Sketch one CNC-friendly part and one print-friendly alternative. Identify inspection surfaces.

[ ] Review your actual printing process/material and what you personally fabricated; record unknowns honestly.

Review notes and reasoning

Selection criteria

Function, stiffness, strength, mass, dimensional stability, environment, fatigue, wear, manufacturing method, quantity, lead time, cost, inspection and repair. Select material plus process plus geometry as a system.

Metals

Aluminium can offer low density and good machinability; steel can provide wear resistance and greater modulus but with higher density. Titanium may suit specific strength/corrosion requirements but has machining/cost challenges. Do not default to titanium because it sounds like F1. Alloy, heat treatment, geometry and loading determine suitability.

CNC design

Consider cutter access, internal corner radii, setup count, datum surfaces, deep narrow pockets, thin-wall distortion, deburring, hole/threads and inspection access. Turning is natural for many axisymmetric parts; milling suits other surfaces. Critical fits may require finishing operations.

Additive manufacturing

FDM extrudes material in layers; SLA uses cured photopolymer; SLS fuses polymer powder. Different processes give different finish, accuracy and mechanical performance. Check orientation, anisotropy, supports or powder removal, wall thickness, build size, shrinkage, inserts, post-processing and inspection. A carbon-filled printed polymer is not equivalent to continuous-fibre laminate CFRP. F1 supplier examples show additive manufacturing used for rapid and detailed tunnel-model production. [S7]

Composites

Fibre-reinforced materials combine reinforcement with a matrix. Continuous carbon laminates can offer high directional stiffness-to-weight; performance depends on fibre direction, lay-up and manufacture. Matrix and interfaces influence shear and delamination behaviour. Consider impact damage, bonding, inserts, drilling, edge distance, inspection and galvanic compatibility. TWI identifies lay-up, cure and void defects as quality concerns. [S8]

Use your real experience

Explain the prototype attachments, modular housing, fastener constraints and fabrication iterations. If you have not performed lay-up, autoclave processing, CNC operation or bonded-joint qualification, say so. Studying materials is useful but not equivalent to hands-on process expertise.

Practice questions - close your notes first

1. Print or machine a sensor mount?

Answer checkpoints: Ask about precision/stiffness/quantity/lead time; compare realistic processes and finish critical locating features as needed.

2. Why not use carbon fibre for every part?

Answer checkpoints: Directionality, joints, geometry, tooling, damage, inspection, lead time and cost; other materials can be better.

3. What is anisotropy?

Answer checkpoints: Properties vary with direction; explain laminate fibre orientation and printed layer orientation as distinct examples.

4. How would you improve a printed bolted housing?

Answer checkpoints: Check load path, layers, inserts/bearing surfaces, wall thickness, creep, fastener preload and repeatability; test rather than assume.

5. What manufacturing issue changed your design?

Answer checkpoints: Use an actual access, available-fastener or instrument-interface example, with your own action and result.

Exit check: Can I explain this simply, sketch it if needed, and answer one follow-up without a script? RED / AMBER / GREEN

9. GD&T, fits, inspection and tolerance stacks

60 minutes | Outcome: Understand function-led tolerancing without pretending to be a GD&T expert.

Actions - produce something, do not just read

[ ] Sketch a bracket with a mounting face, locating feature and bolt pattern. Choose functional datums.

[ ] Mark which features locate and which simply clamp.

[ ] Calculate a simple worst-case gap stack and explain how you would inspect it.

Review notes and reasoning

Key terms

A datum is an ideal reference derived from a physical datum feature. Datums establish how features are located or oriented. Position controls location of axes/centres within a defined tolerance zone. Flatness controls form of a surface without referencing another feature. Perpendicularity controls orientation relative to a datum. ASME resources stress function, manufacture and inspection together. [S9]

Avoid common errors

Do not use flatness to mean parallelism. Do not treat a CAD nominal as a manufacturing guarantee. Do not specify every feature extremely tightly. Use the drawing standard used by the organisation and seek experienced review; ASME and ISO conventions must not be casually mixed.

Fits and location

Clearance fits allow assembly/motion; interference fits create retention through fit and require appropriate analysis. A bolt clearance hole does not necessarily locate a precision part repeatably. Datums, dowels or other locating features may be required. Do not overconstrain an assembly.

Stack example

If gap = opening - part thickness and opening is 20.00 +/-0.10 mm while thickness is 19.50 +/-0.10 mm, the nominal gap is 0.50 mm and worst-case limits are 0.30 to 0.70 mm. Ask if this satisfies function. Statistical stacking needs justified process assumptions, not automatic use.

Inspection

Choose calipers, micrometers, gauges, CMM or other tools suited to the feature and uncertainty needed. Align inspection with drawing datums; consider calibration, access and measurement uncertainty. NPL connects dimensional measurement to part interchangeability and conformance. [S5]

Practice questions - close your notes first

1. What is a datum?

Answer checkpoints: Ideal reference from a real datum feature; explain with the mounting face of your bracket.

2. How do you make a removable sensor return to the same position?

Answer checkpoints: Functional datums/locators, controlled fits, clamp load, cleanliness, wear and inspection.

3. Why not tighten all tolerances?

Answer checkpoints: Cost, lead time and inspection burden; allocate by functional need and stack-up.

4. Explain flatness versus parallelism.

Answer checkpoints: Form of one surface versus orientation relative to a reference. Use a sketch.

Exit check: Can I explain this simply, sketch it if needed, and answer one follow-up without a script? RED / AMBER / GREEN

10. Mechanisms, fasteners, actuation and instrumentation

60 minutes | Outcome: Reason through model systems and safe repeatable operation.

Actions - produce something, do not just read

[ ] Sketch a simple adjustable flap or sensor-position mechanism. Label DOF, stops, bearings, actuator and position feedback.

[ ] Prepare checks for backlash, repeatability, stiffness, travel, cable routing and safe failure.

[ ] Draw a sensor signal chain from measured quantity to reported result.

Review notes and reasoning

Mechanism basics

Define intended degrees of freedom and constrain the others. Check travel, clearance, alignment, friction, backlash, stiffness, wear and overconstraint. Add travel limits and safe fault behaviour where needed. A repeatable angle-setting system is not merely a powerful motor.

Actuator selection

Clarify force/torque through the mechanism, speed, stroke/travel, duty cycle, resolution, backlash, package, environment, feedback and safety. Ideal lever torque is F times perpendicular distance; a real selection includes efficiency, inertia and load cases.

Fasteners

Understand clamping/preload, joint slip/separation, bolt tension/shear, bearing stress, engagement, access and loosening prevention. Torque is an indirect way of producing preload and depends on friction. Do not quote a universal torque-preload constant or add more tightening to fix a weak polymer joint.

Instrumentation

Quantity -> sensor -> conditioning -> acquisition -> calibration/scaling -> analysis -> decision. Check units, range, resolution, bandwidth, drift, noise, mounting and mechanical load path. For unsteady signals, sampling and filtering must suit the dynamics; use anti-alias filtering and adequate sampling, not merely more decimal places.

Pressure tubing and cables

Check leaks, kinks, routing, bend radius, length effects, rubbing, strain relief, connectors and access. Keep instrumentation from changing geometry or carrying unintended mechanical loads. Compare the sensor reading to an independent check before trusting it.

Practice questions - close your notes first

1. A flap angle does not repeat after reversing direction. Why?

Answer checkpoints: Backlash, stiction, compliance, sensor/reference issue or control deadband. Isolate causes with measured tests.

2. Select an actuator for 10 N at a 30 mm lever arm.

Answer checkpoints: Ideal static torque 0.30 Nm; then add real efficiency/dynamics, duty and required margin after clarifying requirements.

3. Why did the load cell show no change?

Answer checkpoints: Check real force transmission, power/excitation, signal conditioning, channel scaling, connections, saturation and independent known load.

4. How would you protect tubing during assembly?

Answer checkpoints: Routing and retention, bend radius, strain relief, labelled connections, access, leak checks and inspection.

Exit check: Can I explain this simply, sketch it if needed, and answer one follow-up without a script? RED / AMBER / GREEN

11. CFD: defend your thesis without turning this into a CFD interview

60 minutes | Outcome: Explain your strongest technical work clearly and with limitations.

Actions - produce something, do not just read

[ ] Prepare a one-minute thesis summary and a three-minute technical version.

[ ] Write the difference between MRF, prescribed sliding mesh and 1DOF dynamic mesh.

[ ] Recheck one mesh table, one timestep table and one experimental comparison before quoting numbers.

Review notes and reasoning

Method distinctions

Your MSc used transient Frame Motion/MRF for performance sweeps, sliding Mesh Motion for changing relative geometry/flow analysis, and dynamic mesh with one rotational DOF for free spin. MRF is often steady in other applications but was not automatically steady in yours. Do not call MRF a physically rotating mesh.

Verification versus validation

Verification asks whether you solve the chosen model adequately; validation compares predictive performance with physical evidence for intended use. Residual reduction alone does not establish either. Check quantities of interest, mass balance, temporal/statistical convergence, mesh, domain and model assumptions.

Your numerical uncertainty

The thesis used four meshes, 3.4 to 14.4 million cells, and a non-monotonic treatment reporting a conservative 19.2% bound for the grid sequence. Do not rebrand this as a tiny formal GCI or claim the finest mesh guarantees truth. Explain why monotonic Richardson assumptions were not justified.

Your experiment comparison

About 163 rpm predicted free spin versus 144 rpm measured is a limited benchmark, not full validation of the power curve. Omitted drivetrain losses plausibly contribute but the discrepancy does not prove their magnitude or exclude other modelling/test errors. The older group locked-rotor drag check was a different, weaker comparison.

Transfer to this role

Understanding sensitive geometry and boundary conditions helps you design test hardware. End answers with the mechanical consequence: accurate position, stable loads, repeatable tests and honest engineering decisions. Keep methods brief unless asked to go deeper.

Practice questions - close your notes first

1. Why choose SST?

Answer checkpoints: Near-wall/adverse-gradient and separation considerations, practical cost and known limitations; do not claim it resolves all turbulence.

2. What would you do if CFD and test differ by 13%?

Answer checkpoints: Check comparability, measurement uncertainty, losses, geometry, numerical/model assumptions; do not label it validated by percentage alone.

3. Why can a converged residual be misleading?

Answer checkpoints: Output drift, inadequate time resolution, conservation or model error can remain.

4. What did the work change in your thinking?

Answer checkpoints: Give a specific evidence-based decision about model fidelity, uncertainty or test design.

Exit check: Can I explain this simply, sketch it if needed, and answer one follow-up without a script? RED / AMBER / GREEN

12. Your project evidence and ownership

45 minutes | Outcome: Have defensible stories without breaching confidentiality or claiming team work as your own.

Actions - produce something, do not just read

[ ] Make an evidence card for MSc thesis, group turbine testing, BEng VAWT and one practical CAD/design task.

[ ] For each write: purpose; my responsibility; tools; one decision; evidence; limitation; lesson.

[ ] Mark confidential details. Prepare a safe generic sketch only if sharing is permitted.

[ ] Verify formal title, dates, marks and module names against records. Do not add MSc Thermodynamics merely because it appeared in an earlier draft.

Review notes and reasoning

Best story order

Lead with a real rig/fixture design for design/manufacturing questions; use MSc for independent analysis, uncertainty and learning; use group testing for teamwork/adaptability; use BEng for design comparisons and MATLAB.

Separate contributions

Say I designed X, I analysed Y, and our team delivered Z. If your colleague did the structural solve, describe your collaboration and interpretation, not personal execution. Being able to describe the whole system does not mean owning all deliverables.

Do not invent success

The group test campaign was constrained by instrumentation problems, generator mismatch and structural failure. Do not claim a completed performance map, full repeatability campaign, formally calibrated validation or guaranteed reliability. A good story explains what was actually achieved and what should change next.

NDA phrasing

I can explain my responsibilities, methods and general lessons, but some geometry and numerical outcomes are confidential. Use only details you are authorised to discuss; previously uploaded thesis material is not automatic permission for employer disclosure.

Evidence template

Problem -> constraints -> options -> my action -> check -> outcome -> limitation -> next improvement. Prefer a specific decision over a long software list.

Practice questions - close your notes first

1. What exactly did you design?

Answer checkpoints: Name a real part and personal contribution, not a broad company programme.

2. What failed in testing and what did you learn?

Answer checkpoints: State the issue neutrally; avoid blame; identify earlier readiness checks and engineering controls.

3. What result are you proud of?

Answer checkpoints: A defensible outcome or better decision, not invented efficiency gains.

4. What can you show under NDA?

Answer checkpoints: Only approved material; offer generalised methods and sketches where permitted.

Exit check: Can I explain this simply, sketch it if needed, and answer one follow-up without a script? RED / AMBER / GREEN

13. General interview and motivation answers

45 minutes | Outcome: Give concise, credible answers and show willingness to collaborate.

Actions - produce something, do not just read

[ ] Record your 90-second introduction and 60-second why-Cadillac answer.

[ ] Prepare four STAR examples: teamwork, setback, deadline and learning/feedback.

[ ] Prepare logistics: start date, location, right to work, salary approach and commitments.

[ ] Write three questions for the interviewers.

Review notes and reasoning

Introduction scaffold

Education -> relevant mechanical/CAD/test contribution -> CFD strength -> why this design/test role. Do not spend the full answer on mesh sizes. End with what you want to contribute and learn.

Motivation scaffold

The role interests me because [real design/test activity]. My evidence is [specific work]. Cadillac interests me because [current verified team observation]. Since the screening I have [actual reading/viewing], and [specific sporting observation] made me more interested.

STAR

Situation and task briefly; actions in first person; honest outcome; reflection. Prepare stories, not memorised paragraphs. For deadlines explain prioritisation, early communication and review; never say you skipped safety or checks to deliver.

Feedback and weakness

Acknowledge your F1 knowledge was developing, explain specific engagement since then, and show genuine curiosity. For a technical weakness choose a real gap such as advanced GD&T or practical composite manufacture and describe a learning plan.

Questions for them

What are typical first-six-month projects? How do model, aero, manufacturing and technicians review work? What proportion is model hardware versus facility systems? How is graduate drawing/release competency developed? What makes a successful first year?

Practice questions - close your notes first

1. Tell me about yourself.

Answer checkpoints: 90 seconds, relevant evidence and role fit; clear communication matters more than a complete biography.

2. Why should we hire you over someone with Formula Student?

Answer checkpoints: Do not criticise others; show your particular analysis-to-test, CAD and learning evidence and acknowledge what you still need to learn.

3. Tell me about disagreement in a team.

Answer checkpoints: Listen, clarify functional criteria, compare evidence, seek review, record decision and support agreed delivery.

4. Where do you want to be in five years?

Answer checkpoints: A genuine capability goal relevant to model/design engineering, not a fabricated lifelong aspiration or promise to leave for CFD.

Exit check: Can I explain this simply, sketch it if needed, and answer one follow-up without a script? RED / AMBER / GREEN

14. Integrated mock interview and final readiness

45 minutes | Outcome: Test performance under pressure, identify gaps and stop passive reading.

Actions - produce something, do not just read

[ ] Run a 30-minute mock: introduction 2 min; motivation/sport 5; project deep dive 7; design scenario 8; behaviour/logistics 5; your questions 3.

[ ] Then use 15 min to score answers and repair the two largest gaps.

[ ] Have a friend interrupt with why, how do you know, what did YOU do and what assumption matters most.

[ ] Before the real interview verify Teams/camera/audio, document permissions, notes, timing and contact details.

Review notes and reasoning

Scenario A: repeatable mount

Design a removable sensor mount on a tight envelope. Clarify loads and position requirement. Sketch a locating strategy, clamp path, manufacturing method and inspection. Discuss cable routing, stiffness, tolerance stack and safe replacement. Do not begin by selecting a fashionable material.

Scenario B: moving model component

An angle-adjustment mechanism drifts in testing. Make safe; confirm the reading and datum; inspect load, backlash, preload, friction, temperature and control feedback. Test causes separately; log and review a controlled change. Check that measurement hardware is not itself the problem.

Scenario C: unexpected aero improvement

A component gives a small force improvement after replacement. Confirm geometry/version, settings, balance zero and baseline repeatability. Inspect mounting gaps/deflection. Compare uncertainty; repeat controlled tests. Do not celebrate a number before excluding a configuration artefact.

Scoring 0–3 per criterion

Score role understanding, sporting interest, project ownership, mechanical reasoning, manufacture/tolerances, verification, teamwork and communication. 0 = absent/wrong; 1 = vague; 2 = sound with some prompting; 3 = clear, specific, honest and independently reasoned. Aim for at least 2 everywhere; the score is a preparation aid, not a hiring prediction.

When you do not know

Say what you know, ask for the missing requirement, state a provisional assumption, reason from first principles and explain how you would verify it. A safe, reviewable approach is better than confident fabrication. No document can guarantee the questions or the result.

Practice questions - close your notes first

1. A senior asks you to release a part without a needed check.

Answer checkpoints: Explain risk and missing check, propose safe alternatives/escalation, follow review authority and change-control rules.

2. You receive a drawing you do not understand.

Answer checkpoints: Clarify standard, datums and function with designer/manufacturing; do not silently reinterpret a critical tolerance.

3. What would you do in your first month?

Answer checkpoints: Learn systems, standards and review process; understand interfaces; complete supervised tasks; seek feedback; verify before release.

4. Are you genuinely ready?

Answer checkpoints: Can you sketch, explain and defend decisions without reading? If not, spend time on active practice rather than another video.

Exit check: Can I explain this simply, sketch it if needed, and answer one follow-up without a script? RED / AMBER / GREEN

Sources and targeted reading

Read selectively. Use official current information for team facts and rules. Some public role pages may move or expire. Keep actual employer documents and confidentiality permissions as the final authority on your experience.

S1 | Cadillac graduate aero-mechanical role, public vacancy

https://haystack.cv/jobs/458753a4-f7d2-4eda-abd1-42fdf9a627d1

Public scope only; actual team allocation is unknown.

S2 | Cadillac official driver announcement

https://www.cadillacf1team.com/news/cadillac-formula-1-r-team-unites-f1-r-stars-to-launch-2026-campaign-bottas

Announced debut drivers; recheck live team details.

S3 | Cadillac official news

https://www.cadillacf1team.com/news

Use for dated current team research; update before interview.

S4 | FIA 2026 operational regulations, issue 11, 1 October 2026

https://www.fia.com/system/files/documents/fia_2026_f1_regulations_-_section_f_operational_-_iss_11_-_2026-10-01.pdf

Check F4 aerodynamic testing provisions and subsequent amendments.

S5 | National Physical Laboratory: measurement and dimensional measurement

https://www.npl.co.uk/products-services/dimensional

Also https://www.npl.co.uk/resources/measurement ; use beginner uncertainty guide.

S6 | Ansys: fundamentals of FEA meshing

https://www.ansys.com/blog/fundamentals-of-fea-meshing-for-structural-analysis

Model ingredients and mesh quality; not a substitute for validated engineering practice.

S7 | 3D Systems / Alpine: wind-tunnel productivity and materials

https://www.3dsystems.com/press-releases/3d-systems-alpine-f1-team-advance-wind-tunnel-productivity-co-developed-material

Example of another team/supplier; not evidence of Cadillac's process.

S8 | TWI: damage inspection in CFRP

https://www.twi-global.com/technical-knowledge/published-papers/phased-array-scanning-of-artificial-and-impact-damage-in-carbon-fibre-reinforced-plastic-cfrp

Composite lay-up, cure, voids and damage/inspection considerations.

S9 | ASME: Principles of GD&T

https://www.asme.org/learning-development/find-course/ippd888-principles-of-gd-t-interpreting-commonly-used-controls

Functional datums and geometric controls; follow employer's applicable standard.

P1 | Isaac Summers, Cranfield MSc individual thesis

User-provided: IndividualThesis_492991_IsaacSummers.pdf

Transient turbine methodology and verification. Verify NDA permission before sharing numerical details.

P2 | Cranfield group turbine report

User-provided: GP__Copy_-2.pdf

Group work: verify personal ownership and actual pressure/FEA load transfer.

P3 | Isaac Summers, BEng VAWT report

User-provided: 21043693-Individual-Project-Report-Backup.pdf

Transient VAWT and MATLAB work. Do not claim helical 3D CFD or a formal timestep convergence study without evidence.

Final reminder

You do not have to sound like a senior F1 designer. Show reliable first-principles thinking, practical design curiosity, truthful ownership, willingness to ask and learn, and enthusiasm for the sport that is actually yours.